Exploratory Data Analysis & Statistical Review

Executive Summary Report: TikTok Claim Classification Project

Overview

The main goal of this project is to analyze the TikTok posts dataset to understand the dynamics between claims and opinions, user engagement performance, and author moderation status. This executive report consolidates key findings from Exploratory Data Analysis (EDA), metrics on variability and virality, the evaluation of missing values and outliers, and the statistical hypothesis test regarding the difference in view counts between verified and unverified accounts.

Problem & Data Structure

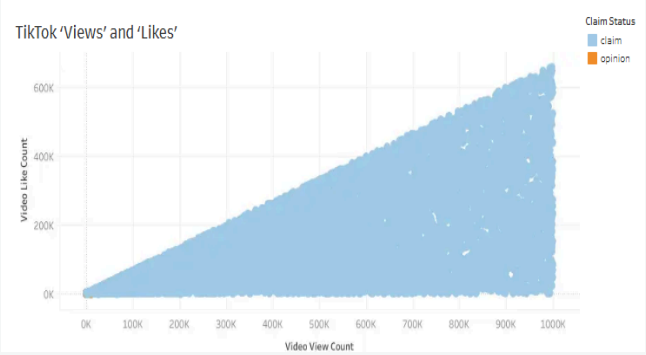
The team identified structural and data quality challenges essential for subsequent predictive modeling:

- **Row and Record Structure:** Each row represents a multimedia post with interaction metrics (views, likes, shares, downloads, comments) and its content classification (claim vs. opinion). Out of a total of 19,382 records, 9,608 (49.57%) are claims and 9,476 are opinions (claims exceed opinions by 132 rows).
- **Null and Missing Values:** Approximately 300 null values were detected, concentrated in **claim_status**, **video_transcription**, and all numerical video interaction variables. The data includes both numeric and objects data types.
- **Highly Skewed Distribution & Outliers:** Engagement metrics exhibit a marked right-skew. On average, interaction metrics contain 3,431 outliers (17.7% of the total data). This reflects the natural behavior of social media where a small percentage of videos goes viral en masse, driving maximum values far away from the mean and the 75th percentile.

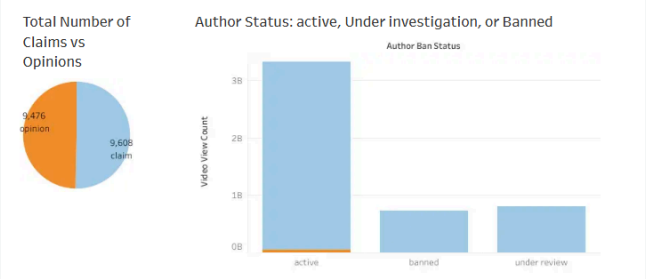
Key Insights (EDA)

EDA & Moderation Findings:

- **Claim vs. Opinion Engagement:** Videos labeled as 'claim' generate dramatically more views, comments, and shares than 'opinion' videos. Claims are the primary traffic driver on the platform.
- **Ban Status & Content:** There is a high concentration of banned or under-review authors among 'claim' videos. The behavior of posting claims exponentially increases the likelihood of receiving a sanction.



- **Banned Author Engagement:** Banned users record nearly double the views, likes, and shares compared to active authors, driven by the high reach of their claim videos.
  See visualizations here.



Hypothesis Testing

A two-sample t-test was conducted to evaluate whether there is a statistically significant difference in view counts (**video_view_count**) between **verified** and **unverified** accounts:

- **Null Hypothesis (H_0):** There is no difference in the number of views between videos posted by verified accounts and unverified accounts ($\mu_1 = \mu_2$).
- **Alternative Hypothesis (H_a):** There is a statistically significant difference in the number of views between videos from verified and unverified accounts ($\mu_1 \neq \mu_2$).
- **Result and Interpretation:** The null hypothesis was rejected with a p-value close to 0 ($p < 0.05$). There is a substantial difference: unverified users receive, on average, more than double the views of verified users. Furthermore, verified accounts show a clear preference for publishing opinion content rather than claims.

Bias Risks & Technical Decisions for the Machine Learning Phase

Outlier Handling (17.7% of the Dataset): Given that virality is inherent to the TikTok ecosystem, outliers should not be removed lightly. It is recommended to use decision tree-based algorithms (Random Forest, XGBoost) that are robust to outliers and skewed distributions, or to apply logarithmic transformations.

Self-Fulfilling / Moderation Bias Risk: If the model learns to classify 'claims' based on author status (**banned** vs. **active**), there is a risk of creating a biased feedback loop. An unverified or high-traffic author could be falsely flagged as a 'claim' or banned simply due to their reach, rather than the actual text or content of the video.

Next Steps

In-Depth Behavioral Analysis: Investigate the behavioral dynamics and content types of unverified users to understand what patterns drive them to outperform verified users by more than double in views.

Isolation of Moderation Variables: Avoid the direct use of **author_ban_status** as the primary predictive feature for claim classification, thereby preventing data leakage and biases in the automated moderation system.

NLP & Tree-Based Modeling Strategy: Prioritize feature extraction from the text (**video_transcription**) combined with engagement metrics using tree-based ensemble models.